

The One- and Two-Alias Ledger at the Natural Counterloop Clock

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Abstract

The finite-radius counterloop bridge is usually stated before its first alias. We compare that alias clock with both relevant weighted bridge cuts. If the counterloop period is tied to the intrinsic endpoint-resolution rank, then the minimal tail-absorbed cut crosses exactly one alias and the original slope-four cut crosses exactly two. The exact counterloop power ledger gives their absolute weighted sizes. Because $\beta R > 1$, the first two impulses grow with explicit powers of $1/\sigma$. This does not prove that the actual typed errors diverge: noisy head or full-trace terms may cancel the shell impulses. It proves that a natural-rank bridge must be alias-inclusive or aggregate, rather than a direct reuse of fixed pre-alias moments.

1 Clock comparison

Let

$$k_\sigma = \frac{L_\sigma}{2 \log \lambda} + O(1), \quad L_\sigma = \log(1/\sigma),$$

be the RH-16 endpoint-resolution clock, and suppose the counterloop period is chosen on this scale. Put

$$h_\sigma = \left\lceil \frac{L_\sigma}{\log(10/7)} \right\rceil, \quad m_\sigma = \lceil 4L_\sigma \rceil.$$

The exact counterloop moment is

$$s_{k,n} = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n), \quad \beta_k \rightarrow \beta = \frac{1}{0.85\sqrt{\lambda}}.$$

2 Alias theorem

Theorem 2.1. *For all sufficiently small σ ,*

$$2k_\sigma < h_\sigma < 4k_\sigma < m_\sigma < 6k_\sigma.$$

Thus the prefix below h_σ contains the first alias order $2k_\sigma$ but not $4k_\sigma$, while the prefix below m_σ contains the first two alias orders but not $6k_\sigma$. At $n = 2\ell k$,

$$\frac{|s_{k,2\ell k}| R^{2\ell k}}{2\ell k} = \frac{1 - k^{-1}}{\ell} (\beta_k R)^{2\ell k}.$$

For $R = 7/5$ and $\ell = 1, 2$, these quantities have logarithmic growth exponents

$$\chi_\ell = \frac{\ell \log(\beta R)}{\log \lambda}, \quad \chi_1 = 0.463406944517003\dots, \quad \chi_2 = 0.926813889034006\dots$$

Proof. The alias orders $2\ell k_\sigma$ have slopes $\ell/\log \lambda$. Direct calculation gives

$$\begin{aligned} \frac{1}{\log \lambda} &= 1.930709\dots < \frac{1}{\log(10/7)} = 2.803673\dots \\ &< \frac{2}{\log \lambda} = 3.861418\dots < 4 < \frac{3}{\log \lambda} = 5.792128\dots \end{aligned}$$

The exact moment formula at a multiple of $2k$ reduces to $(2k-2)\beta_k^{2\ell k}$, proving the weighted identity. Since $\beta_k \rightarrow \beta$ and $k_\sigma = L_\sigma/(2\log \lambda) + O(1)$, taking logarithms and dividing by L_σ gives χ_ℓ . $\square$

3 Protocol and scope

The computation reports all five slopes, the alias counts below each bridge cut, βR , and the first two growth exponents. It also evaluates the exact limiting-radius impulse formula for moderate integer k .

Remark 3.1. RH-16 proves an endpoint singular-value rank, not equality with the noisy modulus-head cardinality. The clock identification in this paper is therefore a typed route audit. Moreover, a large shell term can cancel against a large noisy term; no lower bound for E_σ or D_σ is claimed. Gates A–E remain false/open.